Exp No:28

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COUNTING SEMAPHORE IN SYSTEMC

AIM:

To implement Counting Semaphore allowing multiple resource access.

SOFTWARE REQUIRED:

• Ubuntu (or WSL Ubuntu)

• g++

• CMake

• SystemC

PROGRAM:

```
#include <systemc.h>
SC_MODULE(counting_semaphore) { int
sem; void task() { while(true) { wait(1,
SC_SEC); if(sem > 0) { sem--; cout <<
"Task Accessing Resource at " <<
sc_time_stamp()
<< " Remaining: " << sem << endl;
wait(2, SC_SEC); sem++;
}
}
}
SC_CTOR(counting_semaphore) {
sem = 2;
```

```
SC_THREAD(task);

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}

};

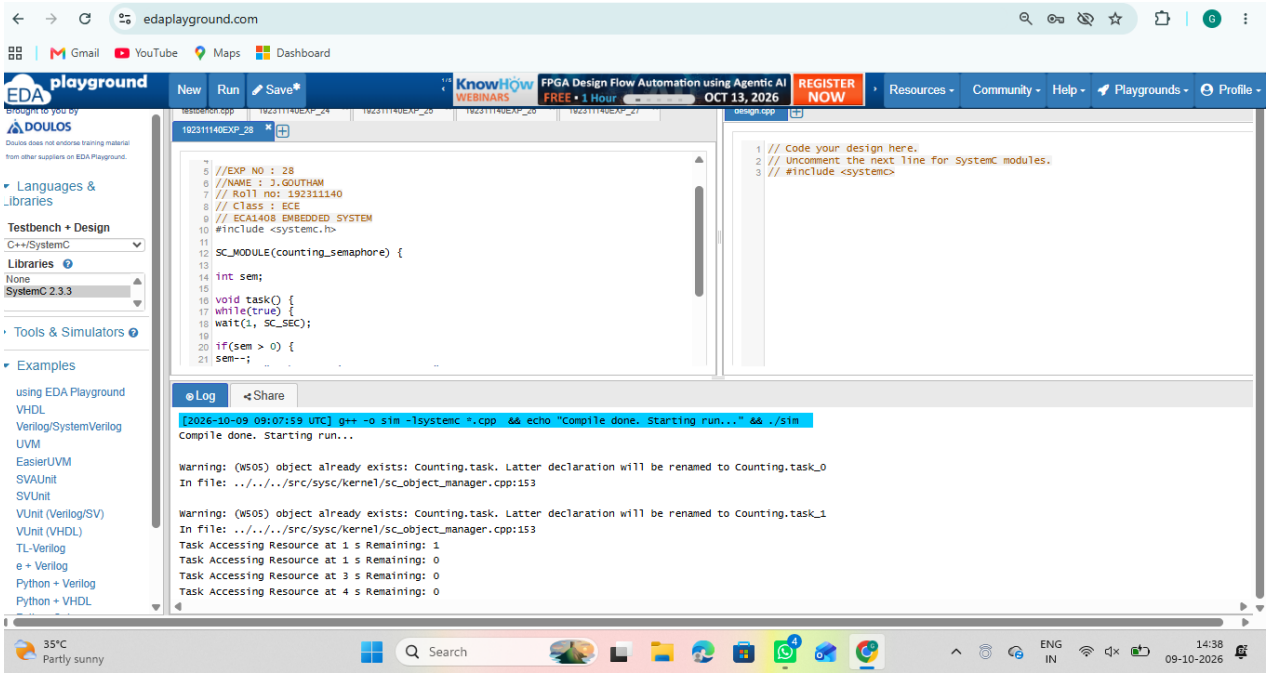
int sc_main(int argc, char* argv[]) {

counting_semaphore obj("Counting");

sc_start(10, SC_SEC); return 0;

}
```

OUTPUT:



RESULT:

Thus the Counting Semaphore was created using SystemC and verified